

TRADING ON PREDICTIONS

One year at a time, and a memory your group can read

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Tue Sep 1, 2026

PRESENTATIONS

Starting Today

- 1 Three groups, drawn at random, at the start of every class
- 2 Five minutes each, on what you got done last session
- 3 Show the repository — the skill, the code, the numbers

What broke is more interesting than what worked, and it is what the rest of the room needs to hear. A presentation that reports only success is a presentation nobody learns from.

What to Cover

What you decided

Which multiple,
which peer group,
what minimum size.
And why.

What surprised you

The company where
the answer was
obviously wrong, and
what it turned out to
be.

What you would change

You have run it now.
What is the first
thing you would do
differently?

FROM A GAP TO A TRADE

You Have a Number for Every Stock

Own P/E divided by peer median P/E. Below one is cheap relative to its peers; above one is expensive.

That is a prediction, whether or not you meant it as one. It says: this company should trade closer to its peers than it does.

The question today is whether that prediction is worth anything.

A threshold

“Buy anything below 0.7.” How many stocks is that? Some years, none. Some years, four hundred.

A sort

“Buy the cheapest tenth.” Always the same number of positions, whatever the market is doing.

Ranking also throws away the scale of the signal, which is the part you trust least. You are betting on the order, not the magnitude.

Sort, Do Not Threshold

Long, Short, or Both

Long only

Buy the cheapest decile. Most of what you earn is just the market.

Long-short

Cheapest decile long, most expensive short. What is left is the signal.

Long-short, dollar neutral

Equal dollars each side. The version we will build.

The long-short version is a much harder test, and it is the one that tells you whether you found anything.

ONE YEAR AT A TIME

The Loop

- 1 Pick a date — the last trading day of June, say
- 2 Compute the signal for every stock, using only what was known that day
- 3 Sort, and form the portfolio
- 4 Hold it for one year. No trading, no rebalancing
- 5 Record the return
- 6 Move forward a year and do it again

Twenty-odd repetitions. That is your entire sample, and it is worth remembering how small a number that is.

Why Once a Year

THE SIGNAL MOVES SLOWLY

It is built from annual filings. Recomputing it monthly gives you eleven months of the same answer and eleven months of trading costs.

AND IT KEEPS THE TEST HONEST

Annual holding periods make the overlapping-returns problem go away. Monthly rebalancing with annual data does not give you twelve times the evidence.

Equal weighted

Same dollars in each name.
Simple, and dominated by small companies — which is where most anomalies live, and where they are hardest to trade.

Value weighted

Weights by market cap. A much more honest test, and usually a much weaker result.

Report both. A strategy that works equal-weighted and dies value-weighted has told you exactly what it is.

Equal or Value Weighted

Build It

TELL GLM

Write a backtest. For each year from 2001 to 2025: on June 30, compute the peer-median signal for every stock using the valuation skill, sort into deciles, go long the cheapest decile and short the most expensive, equal weighted, hold twelve months, and record the return. Plan it before you start.

Then read the plan before you let it write anything.

WHAT WOULD HAVE RUINED IT

The Restatement Problem

MRY is the most recent restatement. It tells you today what nobody knew then.

A company that restated its 2008 earnings in 2011 shows up in **MRY** with the restated number — so a backtest that forms a 2009 portfolio is using information from 2011. The strategy will look better than it was.

Use the as-reported dimensions, **ARY** and **ARQ**, for anything historical.

reportperiod

When the fiscal year ended.
Nobody outside the company
knew the numbers yet.

datekey

When it was filed with the SEC.
This is the first day you could
have acted on it.

Sorting on **reportperiod** gives you a few weeks to a few months of
free information, every year, in every name. It is the single most
common way a backtest in this course goes wrong.

Filed, Not Ended

Who Is in the Sample

Ask a question you cannot answer from the schema: does the data still contain companies that stopped existing?

If it holds only companies listed today, then every 2005 portfolio you form is made of firms you already know survived to 2026. That is not a small bias. It is the difference between a strategy and an illusion.

TELL GLM

Count distinct tickers with data in 2005. Then count how many of those still have data in the most recent year, and show me some that do not.

Three More

The peer median

Computed from the same date, or
from data that arrived later?

Delisting returns

A company that goes to zero has to be
recorded as going to zero

Costs and borrow

The short side is not free, especially in
the small names

None of these will throw an error. Each of them makes the answer
better than the truth.

READING THE RESULT

Average return

The mean of your annual long-short returns.

Standard deviation

How much those returns bounced around, year to year.

The ratio, and its t-stat

With twenty-five observations, the t-stat is doing a lot of work.

Twenty-five annual returns is a small sample. A strategy earning four percent a year with fifteen percent volatility is not distinguishable from nothing.

Three Numbers

Compare It to Something

Four percent a year is a fact about your strategy. It is not yet a finding.

The market

Did the long side simply do what stocks did?

A random sort

Run it with the signal shuffled. What does noise earn in your backtest?

Value itself

Plain P/E, no peer group. Did the peer adjustment add anything?

The third comparison is the real test of last session's work.

Look at the Years

TELL GLM

Plot the annual return of the long-short portfolio, year by year. Mark the best and worst years and tell me what the portfolio held in each.

An average built out of one enormous year and twenty-four small negative ones is a different animal from a steady four percent, and the summary statistics will not tell you which you have.

A MEMORY YOUR GROUP CAN READ

Three People, No Shared Memory

The conversation you had with GLM on Tuesday is invisible to your groupmates, and gone to you.

The code survives. The reasoning does not — why you used `ARY`, which peer definition you rejected, what you tried on the short side that did not work. That is exactly what the next person needs, and exactly what nobody writes down.

What a Memory File Is

One file per session

Dated, in the repository, next to the code it describes.

Written by GLM

At the end of the session, from the conversation it just had. You correct it.

Read by GLM

At the start of the next session, before it touches anything.

It serves your groupmates and it serves the model. Both start every session knowing nothing.

- 1 Date, and who was working
- 2 What we set out to do
- 3 What we decided, and what we rejected — with the reason
- 4 What broke, and whether it is fixed
- 5 What is unfinished, and where to pick it up

Point 3 is the one that matters and the one that gets dropped. A decision without its reason gets re-litigated in two weeks.

What Goes in It

Build the Skill

TELL GLM

Make a skill that writes a session memory file. It should trigger when I say we are wrapping up. Write to `memory/YYYY-MM-DD.md` with those five headings, drawn from this conversation. Keep it under a page.

Then a second one, or a line in `CLAUDE.md` :

At the start of every session, read the three most recent files in `memory/`.

Try It Now

- 1 Build the skill
- 2 Say: we are wrapping up
- 3 Read what it wrote — and fix what it got wrong about your own afternoon
- 4 Commit and push, so the other two can read it before Thursday

Step 3 is not optional. A memory file nobody checked is a confident record of a session that did not happen.

BEFORE THURSDAY

Three Things

- 1 Get the backtest running end to end, however crude
- 2 Re-run it with `ARY` and `datekey`, and record what changed
- 3 Write the memory file, and push it

Thursday we stop borrowing the peer median and start predicting the multiple — which turns valuation into a regression, and gives you something to compare today's result against.